

Finite Difference Valuation of Autocallable Notes Linked to Apple

1 Executive Summary

This report presents the valuation of an Autocallable Structured Note issued by Morgan Stanley Finance LLC, linked to Apple Inc. (AAPL). We implement two finite difference pricing frameworks in Python, explicit and Crank-Nicolson scheme, to evaluate the note's payoff structure under several volatility scenarios. The fair value from our model is approximately \$973.73, compared to a market price of \$978.50, implying a deviation of about -0.49% .

2 Data and Product Description

All market inputs were sourced from Bloomberg using the Bloomberg Terminal.

- **Interest Rate**

Using SWDF command in Bloomberg, we extract discount rate starting on December 9, 2024 (pricing date) from OIS term structure. Note that following market convention, the interest rates are quoted on December 5, 2024. Using linear interpolation, we get the discount rate through January 14, 2026 (maturity date) is , the interest rate converted from this value is risk-free rate.

Table 1: Discount rate of OIS

	Dec. 9, 2025	Jan. 14, 2026	Jun. 9, 2026
Discount rate	0.959527	0.956028	0.941839
Risk-free rate	4.13148%	4.09308%	3.99838%

- **Dividend Yields**

Using OVME command in Bloomberg, we get dividend yield for three stocks between December 9, 2024 (pricing date) and January 14, 2026 (maturity date), which is actually calculated by method from OPDF command in Bloomberg.

$$q = 0.403\%.$$

- **Implied Volatilities**

Using OVM command in Bloomberg, we get volatility surface of stocks, we interpolate the potential volatilities on the coupon dates, exporting barrier and ATM volatilities for both average and terminal maturities as follows:

Table 2: Volatility Inputs

Scenario	Volatility (σ)
Average Barrier Volatility	0.2909
Terminal Barrier Volatility	0.2737
Terminal ATM Volatility	0.2361
Average ATM Volatility	0.2212

- **Structured Note Terms:** Sourced directly from the offering document (Morgan Stanley), with validation against Bloomberg fields.

The key parameters of the note are summarized in Table 3.

Table 3: Product Parameters

Parameter	Value
Underlying	Apple Inc. stock
Initial Price (S_0)	\$246.75
Knock-In Level	\$192.465 (78.0% of S_0)
Coupon Barrier	\$192.465 (78.0% of S_0)
Principal (N)	\$1,000
Coupon Rate	0.75% per month
Equity Ratio	4.05268
Maturity (T)	401 days $\approx$ 1.0986 years
Risk-Free Rate (r)	4.04%
Dividend Yield (q)	0.403%
Market Price	\$978.50
Series of Valuation Dates	See original sheet

3 Algorithm

We built a finite difference grid that time grid has $i_{\max} = n_i \cdot T$ steps, where n_i is the number of sub-steps per day and T is the number of days between the pricing date and the maturity date, the stock grid spans $[SL, SU]$ with $j_{\max}$ divisions, centered at S_0 . We perform the Black-Scholes PDE:

$$\frac{\partial V}{\partial t} + (r - q)S \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV$$

where:

- r is the continuous risk-free rate.
- q is the continuous dividend rate.
- σ is the volatility of underlying stock.
- $\Delta S = \frac{SU - SL}{j_{\max}}$ is the length of stock step.
- $\Delta t = \frac{1}{365 * n_i}$ is the length of time step.

3.1 Contingent Coupon

A contingent coupon at an monthly rate of 0.75% will be paid on each coupon payment date but only if the closing value of the underlying on the immediately preceding valuation date is greater than or equal to the coupon barrier value:

$$CPN_j^{ic} = N \cdot 0.75\% \cdot \mathbf{1}_{\{S_j^{ic} \geq \text{Coupon Barrier}\}}$$

where ic is time step on coupon payment dates.

3.2 Autocall feature

If the time step corresponds to an autocall date and the stock is greater than or equal to the initial price, the security will be automatically called on the immediately following coupon payment date for:

$$V_j^{ic} = N + CPN_j^{ic}.$$

3.3 Terminal Condition (Payment at Maturity)

At maturity, payment depends on whether there is a knock-in event. A knock-in event will occur if, on any scheduled trading day during the observation period, the closing value of the underlying is less than the knock-in value.

If the security doesn't redeemed before the maturity and a knock-in event has occurred:

$$V_j^{imax} = CPN_j^{imax} + \begin{cases} N, & \text{if } S_j^{imax} \geq S_0 \\ \text{Equity Ratio} \cdot S_j^{imax}, & \text{otherwise} \end{cases}.$$

and if a knock-in event hasn't occurred:

$$V_j^{imax} = CPN_j^{imax} + N$$

3.4 Boundary Condition

1. Upper Boundary:

During the autocall period, the upper boundary condition is the discounted redemption value on the following coupon date. And prior to the first autocall date, it is the sum of the discounted redemption value on the coupon date corresponding to the first autocall date and all discounted contingent coupon between that day and the first autocall date:

$$V_{jmax}^i = \begin{cases} \sum_{i \leq ic < iac_1} CPN_{jmax}^{ic} \cdot e^{-r(ic-i)} + (N + CPN_{jmax}^{iac_1}) \cdot e^{-r(iac_1-i)}, & \text{if } i < iac_1 \\ (N + CPN_{jmax}^{ic}) \cdot e^{-r(ic-i)}, & \text{otherwise} \end{cases}.$$

where iac_1 is the following coupon date for the first autocall date, and ic is the following coupon date after the time step i .

This upper boundary condition applies to both grids with and without a knock-in event.

2. Lower Boundary:

For the grid with a knock-in event, we set $SL = 0$, and therefore

$$VT_0^i = 0.$$

For the grid without a knock-in event, we set $SL = 0.78 \cdot S_0$, which is the knock-in level, leaving the grid where the stock price is lower than that price unsolved, and

$$VN_0^i = VT_{jb}^i$$

where jb satisfies $S_{jb} = 0.78 \cdot S_0$, which is the knock-in level in the grid with a knock-in event.

3.5 Finite Difference

We implemented two finite difference methods for pricing the structured note:

1. Explicit Scheme:

The prices are updated following a backward recursion:

$$V_j^i = A_j V_{j+1}^{i+1} + B_j V_j^{i+1} + C_j V_{j-1}^{i+1}$$

where

$$\begin{aligned} A_j &= \left(\frac{1}{2} \sigma^2 (SL/\Delta S + j)^2 + \frac{(r-q)(SL/\Delta S + j)}{2} \right) \Delta t, \\ B_j &= 1 - \sigma^2 (SL/\Delta S + j)^2 \Delta t, \\ C_j &= \left(\frac{1}{2} \sigma^2 (SL/\Delta S + j)^2 - \frac{(r-q)(SL/\Delta S + j)}{2} \right) \Delta t. \end{aligned}$$

Specifically, on coupon date, i.e. $i = ic$,

$$V_j^{ic} \leftarrow V_j^{ic} + CPN_j^{ic}$$

and to handle the autocall feature, on coupon date during the autocall period, for $j \geq j_0$, where j_0 satisfies $S_{j_0} = S_0$, we correct value on both grids:

$$V_j^{ic} = N + CPN_j^{ic}$$

2. Crank-Nicolson Scheme:

The prices are updated following:

$$-A_j V_{j-1}^i + (1 - B_j) V_j^i - C_j V_{j+1}^i = A_j V_{j-1}^{i+1} + (1 + B_j) V_j^{i+1} + C_j V_{j+1}^{i+1}$$

where

$$\begin{aligned} A_j &= \frac{1}{4} (\sigma^2 (SL/\Delta S + j)^2 + (r-q)(SL/\Delta S + j)) \Delta t, \\ B_j &= -\frac{1}{2} (\sigma^2 (SL/\Delta S + j)^2 + r) \Delta t, \\ C_j &= \frac{1}{4} (\sigma^2 (SL/\Delta S + j)^2 - (r-q)(SL/\Delta S + j)) \Delta t. \end{aligned}$$

Specifically, the value V_j^i depends on all values at the next time step V_*^{i+1} and the current time step V_*^i , as a consequence, the error in any V_*^i can propagate and be amplified by errors in all other V_*^i . If we just correct the value V_j^{ic} for $j \geq j_0$ on the coupon date during the autocall period locally, we introduce inconsistency in the grid which leads to potentially explosive

numerical instability. In other words, in the Crank-Nicolson method, we only solve the grids below j_0 , so the certain accuracy above j_0 will be preserved:

$$V_j^{ic} = \begin{cases} N + CPN_j^{ic}, & \text{if } j \geq j_0 \text{ and stay unsolved} \\ f(A_j, B_j, C_j, V_{j-1}^i, V_{j+1}^i, V_{j-1}^{i+1}, V_j^{i+1}, V_{j+1}^{i+1}) + CPN_j^{ic}, & \text{otherwise} \end{cases}.$$

The grid values are calculated in matrix form at one time,

$$M_2 V^i + b_2 = M_1 V^{i+1} + b_1$$

where

$$M1 = \begin{bmatrix} 1+B_1 & C_1 & 0 & \cdots & 0 \\ A_2 & 1+B_2 & B_2 & \cdots & 0 \\ 0 & A_3 & 1+B_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \ddots & A_{j'-2} & 1+B_{j'-2} & C_{j'-2} \\ 0 & \cdots & 0 & A_{j'-1} & 1+B_{j'-1} \end{bmatrix},$$

$$M2 = \begin{bmatrix} 1-B_1 & -C_1 & 0 & \cdots & 0 \\ -A_2 & 1-B_2 & -C_2 & \cdots & 0 \\ 0 & -A_3 & 1-B_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \ddots & -A_{j'-2} & 1-B_{j'-2} & C_{j'-2} \\ 0 & \cdots & 0 & -A_{j'-1} & 1-B_{j'-1} \end{bmatrix},$$

$$b_1 = \begin{bmatrix} A_1 V_0^{i+1} \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ C_{j'-1} V_{j'}^{i+1} \end{bmatrix}, \quad b_2 = \begin{bmatrix} -A_1 V_0^i \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ -C_{j'-1} V_{j'}^i \end{bmatrix}, \quad \text{and } V^i = \begin{bmatrix} V_1^i \\ V_2^i \\ V_3^i \\ \vdots \\ V_{j'-3}^i \\ V_{j'-2}^i \\ V_{j'-1}^i \end{bmatrix},$$

where $j' = \begin{cases} j_0, & \text{if } i \in \{ic_{ac}\} \\ j_{max}, & \text{otherwise} \end{cases}$, $\{ic_{ac}\}$ are the coupon dates corresponding to autocall dates.

And $j = 0$ satisfies $S_0 = \begin{cases} 0, & \text{in the grid with knock-in event} \\ S_{jb}, & \text{in the grid without knock-in event} \end{cases}$.

The system is solved efficiently using LU factorization for banded matrix.

4 Discussion

4.1 Choice of Volatility

The results in two schemes under different volatility assumptions are summarized in Figure 1. Generally, if the volatility is too small, the probability of hitting coupon barrier will be low, the security always pays coupon, and the trigger event is less likely to occur, resulting in an overvaluation, and if the volatility is too large, the probability of hitting coupon barrier will be too high, the security seldom pays coupon, and the trigger event is more likely to occur, resulting in an undervaluation.

4.2 Comparison of Explicit vs CNFD

- The **Explicit FD method** is easier to implement, but conditionally stable. Careful choice of time and space steps is required to satisfy stability constraints.
- The **CNFD method** is unconditionally stable, produces smoother results, and is less sensitive to grid choices, though computationally more demanding.
- Since both approaches converge to similar results (975.24 vs 973.93), we conclude that either scheme can be used reliably, but CNFD is preferred for robustness, while Explicit FD is advantageous for speed and transparency.

4.3 Numerical Stability and Convergence

To validate convergence, we performed sensitivity checks on the grid size and the choice of the upper boundary S_U . Both tests confirm that the Explicit FD results stabilize around the final price of ~ 975 as we increase the grid size and upper bound stock price limitation.

The grid sensitivity analysis indicates that pricing accuracy is primarily driven by the number of space steps, while the number of time steps plays a secondary role. This outcome is consistent with the intuition that the partial differential equation (PDE) pricing scheme is dominated by the spatial discretization of the diffusion term, whereas the temporal discretization mainly serves to stabilize the scheme.

The experiment on the stock price upper boundary demonstrates that, for this product, setting the upper limit to twice the initial stock price ($2.0 \cdot S_0$) is already sufficient to achieve stable and accurate valuations. Extending the grid further toward higher stock levels does not significantly alter the computed price, confirming that the contract's payoff structure is insensitive to extreme upward movements beyond this range.

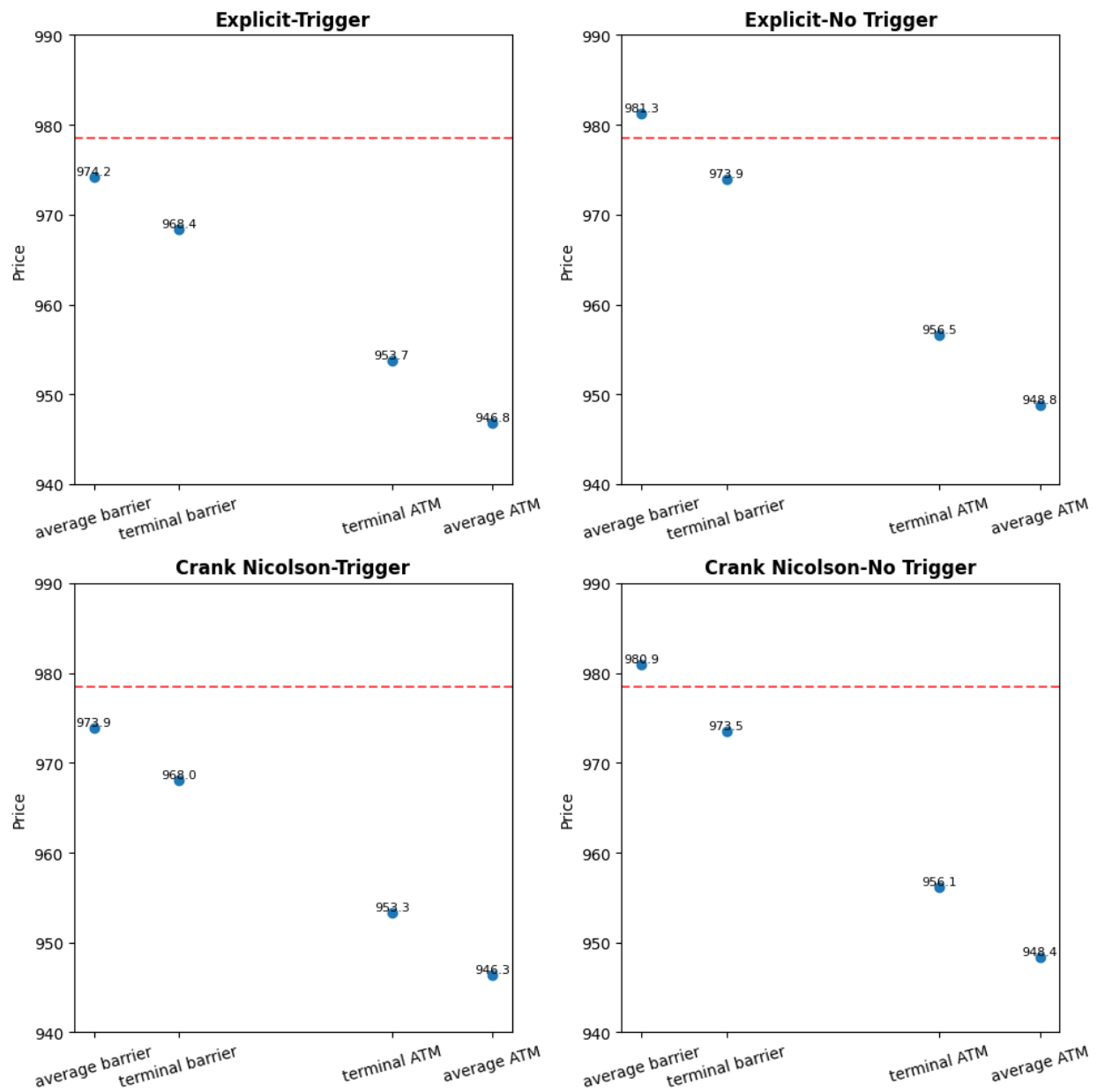


Figure 1: Sensitivity to Volatility

3D Sensitivity of ExplicitFD_noTrigger to Grid Size

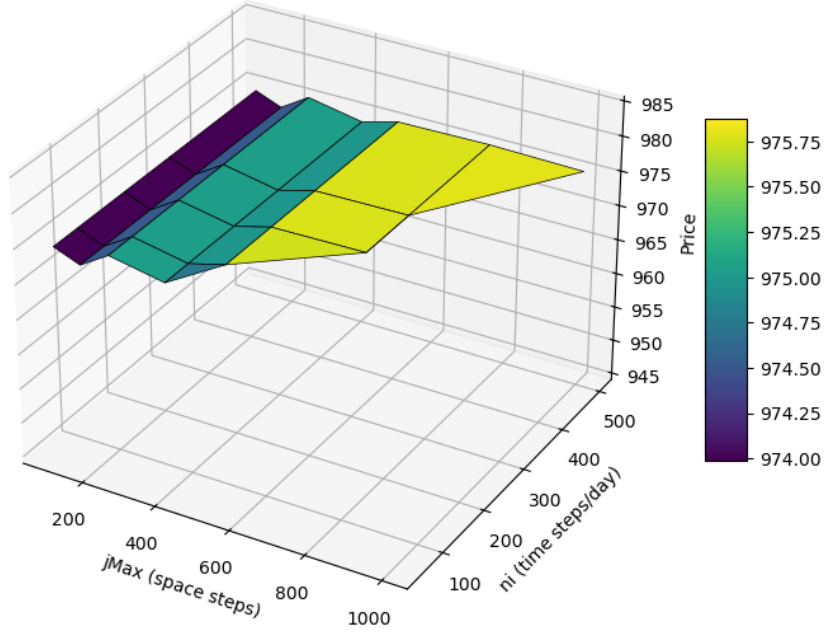


Figure 2: Convergence with respect to spatial grid size $jMax$ (Explicit FD).

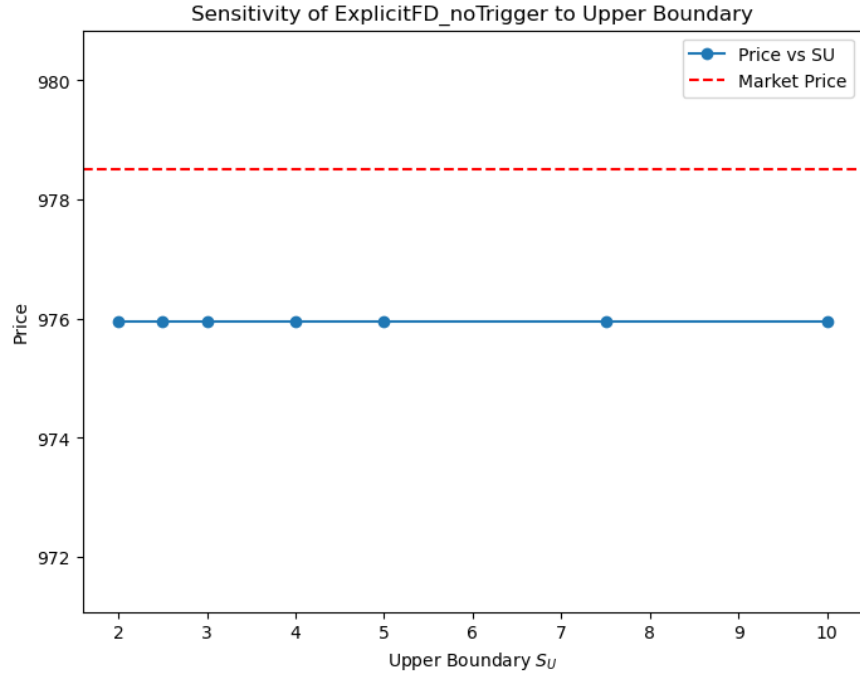


Figure 3: Convergence with respect to upper boundary S_U (Explicit FD).

The plots illustrate that once the grid is sufficiently fine and the boundary large enough, the price stabilizes close to the market benchmark, supporting the validity of our numerical method.

5 Conclusion

The estimation is sensitive to the volatilities, but once the grid is well defined, both methods compute consistent estimation, validating our implementation and confirming the robustness of the pricing framework.

We report results from both methods setting $SL = 0, SU = 2.5S_0, ni = 100, j_{max} = 250$ for grid with trigger, $SL = 0.78S_0, SU = 2.5S_0, ni = 100, j_{max} = 175$ for grid without trigger, and the ATM volatility at the maturity date.

- **Explicit FD (no trigger):**

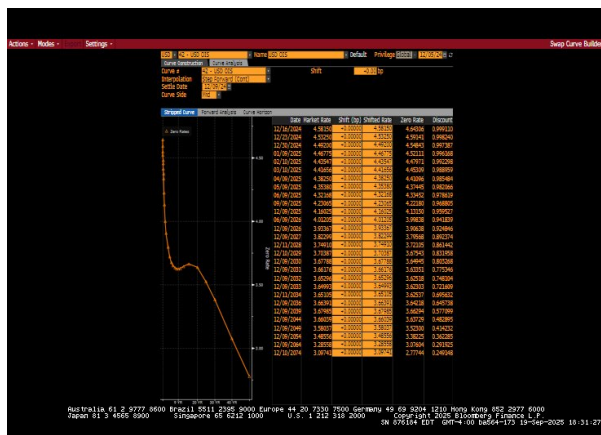
$$P_{\text{ATM}}^{\text{Explicit}} = 973.92,$$

- **Crank–Nicolson FD (no trigger):**

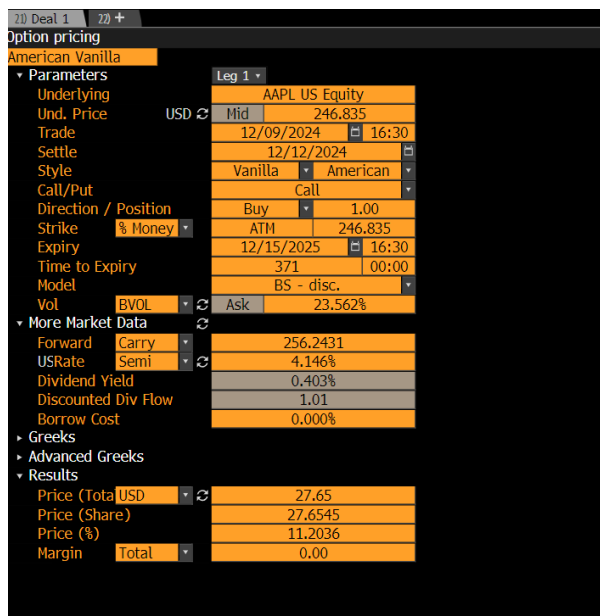
$$P_{\text{ATM}}^{\text{CNFD}} = 973.73,$$

Both are close to the market price, differing by less than 0.5% from the observed value.

Appendix: Data from Bloomberg



(a) OIS



(b) Dividend



(c) Volatility1



(d) Volatility2

Figure 4: Bloomberg Data: OIS, Dividend, and Volatilities